

Two-Way Factorial Designs

STAT452

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Motivation & Factorial Structure

Where are we in the course?

- ▶ We have studied designs with **one** treatment factor: the CRD (one-way ANOVA) and the RCBD (one treatment factor + a blocking factor).
- ▶ Very often we want to study **two or more** treatment factors *at the same time*, and — crucially — how they **interact**.
- ▶ Today: the **two-way factorial design**, where the treatments are all combinations of the levels of two factors.

Remark — the new idea is interaction

A blocking factor is a nuisance we remove. Here *both* factors are of interest, and the central new question is whether the effect of one factor **depends on** the level of the other. That is what an RCBD cannot address.

Example: Sprouting Barley (Oehlert, p.166)

Brewer's malt is made from germinating barley, so brewers want to know the best conditions to germinate it.

- ▶ 30 lots of barley seeds (100 seeds per lot) are randomly split into 10 groups of 3 lots.
- ▶ Each group receives one **treatment**, a combination of:
 - ▶ **water amount** used in germination — 4 ml or 8 ml;
 - ▶ **age of seeds** (weeks after harvest) — 1, 3, 6, 9, 12.
- ▶ Response: number of seeds (out of 100) that germinated.

This is a 2×5 **factorial design** with $n = 3$ replicates per cell (30 observations in total).

The Sprouting Barley data

	Age of seeds (weeks)				
	1	3	6	9	12
Water					
4 ml	11	7	9	13	20
	9	16	19	35	37
	6	17	35	28	45
8 ml	8	1	5	1	11
	3	7	9	10	15
	3	3	9	9	25

- ▶ Each cell contains $n = 3$ replicate lots.
- ▶ We will use this data set throughout the lecture and analyse it in R at the end.

Basic terminology

The barley experiment has 10 treatments with a **factorial structure**.

Factor an experimentally adjustable variable, e.g. water amount, age of seeds.

Level a possible value of a factor: water has 2 levels (4, 8 ml); age has 5 levels (1, 3, 6, 9, 12 weeks).

Treatment a **combination** of factor levels. Here the $2 \times 5 = 10$ treatments are

(4, 1), (4, 3), (4, 6), (4, 9), (4, 12), (8, 1), (8, 3), (8, 6), (8, 9), (8, 12).

Remark

A two-way factorial design is, at heart, still a **CRD with ab treatments**. The **factorial structure** of those treatments is the extra ingredient we will exploit.

Full k -way factorial designs

Definition — full factorial design

With k factors having $L_1, L_2, \dots, L_k$ levels, a **full k -way factorial design** uses all $L_1 \times L_2 \times \dots \times L_k$ combinations as treatments.

Balanced every treatment group has the **same** number of replicates; otherwise the design is **unbalanced**.

- ▶ **Question.** How many units are there in a 3×2 design with 4 replicates?
 $\Rightarrow 3 \times 2 \times 4 = 24$.
- ▶ Balance brings big computational advantages (orthogonality), but is not always achievable: a dropped test tube can unbalance an originally balanced design.

Data Layout & the Means Model

Data for a two-way $a \times b$ design with n replicates

	B -level 1	B -level 2	$\dots$	B -level b
A -level 1	$y_{111}, \dots, y_{11n}$	$y_{121}, \dots, y_{12n}$	$\dots$	$y_{1b1}, \dots, y_{1bn}$
A -level 2	$y_{211}, \dots, y_{21n}$	$y_{221}, \dots, y_{22n}$	$\dots$	$y_{2b1}, \dots, y_{2bn}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
A -level a	$y_{a11}, \dots, y_{a1n}$	$y_{a21}, \dots, y_{a2n}$	$\dots$	$y_{ab1}, \dots, y_{abn}$

- ▶ y_{ijk} = the k -th replicate in the cell (level i of A , level j of B).
- ▶ Indices: $i = 1, \dots, a$; $j = 1, \dots, b$; $k = 1, \dots, n$.
- ▶ Total number of observations: $N = abn$.

The means model

Definition — means (cell-means) model

$$y_{ijk} = \mu_{ij} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2),$$

with $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, n$.

- ▶ μ_{ij} = mean response in the cell (level i of A , level j of B).
- ▶ There are ab cell means — one free parameter per cell.

Remark

The means model treats the design as a **CRD with ab treatments** and *ignores* the factorial structure. It is the most flexible model (it can fit any pattern of cell means), and every other model we write today is **nested** inside it.

Interaction: Definition & Contrasts

The two-way interaction contrast

For two levels (i_1, i_2) of factor A and two levels (j_1, j_2) of factor B , define the **interaction contrast**

$$C = \mu_{i_1 j_1} - \mu_{i_1 j_2} - \mu_{i_2 j_1} + \mu_{i_2 j_2}.$$

It has **two equivalent readings**:

$$C = \underbrace{(\mu_{i_1 j_1} - \mu_{i_1 j_2})}_{B\text{-effect when } A=i_1} - \underbrace{(\mu_{i_2 j_1} - \mu_{i_2 j_2})}_{B\text{-effect when } A=i_2}$$

$$C = \underbrace{(\mu_{i_1 j_1} - \mu_{i_2 j_1})}_{A\text{-effect when } B=j_1} - \underbrace{(\mu_{i_1 j_2} - \mu_{i_2 j_2})}_{A\text{-effect when } B=j_2}$$

C compares the effect of one factor *across* two levels of the other.

No interaction — the definition

Definition — no interaction

Factors A and B have **no interaction** if and only if every interaction contrast is zero:

$$\mu_{i_1 j_1} - \mu_{i_1 j_2} - \mu_{i_2 j_1} + \mu_{i_2 j_2} = 0 \quad \text{for all } i_1, i_2, j_1, j_2.$$

Two equivalent interpretations:

- ▶ the effect of A on Y **does not change** with the level of B ; and
- ▶ the effect of B on Y **does not change** with the level of A .

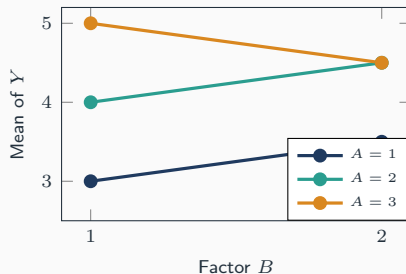
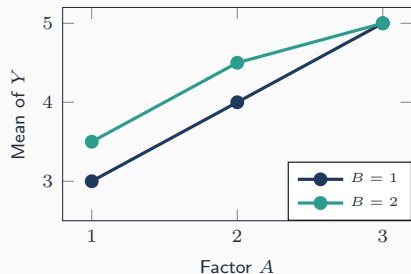
Caution

Conversely, A and B **interact** if the effect of A changes with the level of B (equivalently, the effect of B changes with the level of A). Interaction is a *symmetric* property of the two factors.

Interaction Plots

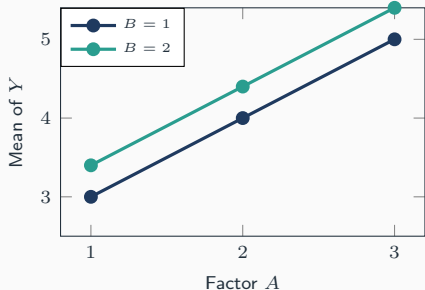
Interaction plots

Plot the cell means μ_{ij} against the levels of one factor, with a separate line for each level of the other (the **trace** factor).



- The two plots (swapping which factor is on the x -axis) carry the **same** information.

Parallel lines $\Leftrightarrow$ no interaction



- ▶ When the lines are **parallel**, the gap between them (the B -effect) is the *same* at every level of A — and the slope (the A -effect) is the same at every level of B .
- ▶ Exactly parallel lines $\Leftrightarrow$ all interaction contrasts = 0 $\Leftrightarrow$ **no interaction**.

Reading interaction plots — a checklist

Given the two profiles below (parallel, horizontal lines with a gap or a slope):

- ▶ No interaction whenever the lines are parallel.
- ▶ Factor A (on the x -axis) has an effect if the lines are *not horizontal* (they slope).
- ▶ Factor B (the trace) has an effect if there is a *vertical gap* between the lines.

Remark — four qualitative cases

Parallel + sloped + gap: both main effects, no interaction.

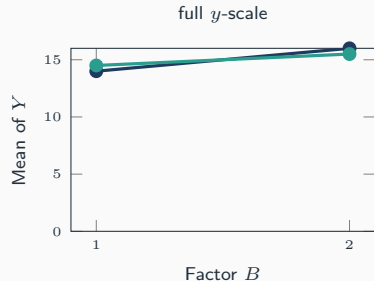
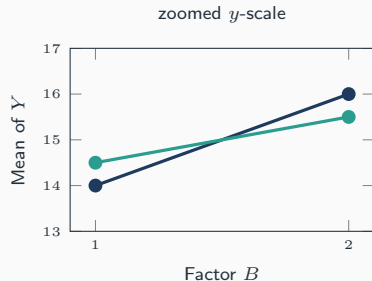
Parallel + sloped + no gap: only A matters.

Parallel + flat + gap: only B matters.

Non-parallel: interaction present (main-effect statements need care).

Beware: the Y -scale distorts “parallel”

The visual impression of “parallel or not” depends on the vertical scale.



Caution

Always check the Y -scale and ask whether the change in slopes is **large enough to matter** in practice. A formal F -test for interaction (later) settles the question objectively.

The Additive (Main-Effects) Model

The additive model

Definition — additive (main-effects) model

$$y_{ijk} = \mu + \underbrace{\alpha_i}_{A \text{ main effect}} + \underbrace{\beta_j}_{B \text{ main effect}} + \varepsilon_{ijk}.$$

- ▶ It **does** take the factorial structure into account (separate effects for A and B), unlike the means model.
- ▶ It is **nested** in the means model $y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$: the means model reduces to the additive model exactly when

$$\mu_{ij} = \mu + \alpha_i + \beta_j \quad \text{for all } i, j.$$

The additive model assumes no interaction

If the additive model holds, $\mu_{ij} = \mu + \alpha_i + \beta_j$, then every interaction contrast collapses:

$$\begin{aligned}\mu_{i_1j_1} - \mu_{i_1j_2} - \mu_{i_2j_1} + \mu_{i_2j_2} &= (\mu + \alpha_{i_1} + \beta_{j_1}) - (\mu + \alpha_{i_1} + \beta_{j_2}) \\ &\quad - (\mu + \alpha_{i_2} + \beta_{j_1}) + (\mu + \alpha_{i_2} + \beta_{j_2}) = 0.\end{aligned}$$

Caution — the converse matters

So “additive model” $\Rightarrow$ no interaction. But real data follow the *means* model, under which the interaction contrasts need not be zero. If we fit the additive model when interaction is truly present, that interaction is wrongly absorbed into the error — inflating MSE and biasing the tests.

The Main-Effects-Interaction Model

The main-effects-interaction model

Definition — full (main-effects-interaction) model

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}.$$

- ▶ $(\alpha\beta)_{ij}$ is a **single parameter**, *not* a product: $(\alpha\beta)_{ij} \neq \alpha_i \times \beta_j$.
- ▶ In matrix form the cell-mean array decomposes as

$$[\mu_{ij}] = \mu \mathbf{1}\mathbf{1}^\top + [\alpha_i]\mathbf{1}^\top + \mathbf{1}[\beta_j]^\top + [(\alpha\beta)_{ij}].$$

Remark

This model is **equivalent** to the means model: same fitted values, residuals, and SSE. It simply *re-expresses* the *ab* cell means as an overall level, two sets of main effects, and an interaction array.

The full model is over-parameterized

Counting parameters in the full model:

$$\underbrace{1}_{\mu} + \underbrace{a}_{\alpha_i} + \underbrace{b}_{\beta_j} + \underbrace{ab}_{(\alpha\beta)_{ij}} = 1 + a + b + ab.$$

But the means model has only ab free parameters (the cell means).

- ▶ Two equivalent models must have the **same** number of free parameters.
- ▶ Hence the full model is **over-parameterized**: its parameters are *not* uniquely determined unless we impose **constraints**.
- ▶ The redundancy count is $(1 + a + b + ab) - ab = 1 + a + b$ — exactly the number of constraints we must add.

Identifiability & Constraints

Baseline (reference) constraints — R's default

R sets all parameters at the **first level** of each factor/interaction to zero:

$$\alpha_1 = 0, \quad \beta_1 = 0, \quad (\alpha\beta)_{1j} = (\alpha\beta)_{i1} = 0 \quad \text{for all } i, j.$$

Effectively this leaves

$$\underbrace{1}_{\mu} + \underbrace{(a-1)}_{\alpha} + \underbrace{(b-1)}_{\beta} + \underbrace{(a-1)(b-1)}_{(\alpha\beta)} = ab$$

free parameters — the same as the means model. Interpretation (baseline coding):

- ▶ $\mu = \mu_{11}$ (the baseline cell);
- ▶ $\alpha_i = \mu_{i1} - \mu_{11}$: effect of moving A from level 1 to i , at $B = 1$;
- ▶ $\beta_j = \mu_{1j} - \mu_{11}$: effect of moving B from level 1 to j , at $A = 1$;
- ▶ $(\alpha\beta)_{ij} = \mu_{ij} - \mu_{i1} - \mu_{1j} + \mu_{11}$.

Zero-sum constraints — preferred for factorial data

Definition — zero-sum constraints

$$\sum_{i=1}^a \alpha_i = 0, \quad \sum_{j=1}^b \beta_j = 0, \quad \sum_{i=1}^a (\alpha\beta)_{ij} = 0 \quad \forall j, \quad \sum_{j=1}^b (\alpha\beta)_{ij} = 0 \quad \forall i.$$

Every row sum and column sum of the array $\{(\alpha\beta)_{ij}\}$ is zero.

Parameter count again reduces to ab :

$$1 + (a - 1) + (b - 1) + (a - 1)(b - 1) = ab.$$

- $\alpha_a = -\sum_{i < a} \alpha_i$, $\beta_b = -\sum_{j < b} \beta_j$; the last row and column of $\{(\alpha\beta)_{ij}\}$ are fixed by the constraints.

Deriving the meaning of the zero-sum parameters

Start from $\mu_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}$ and sum over i :

$$\sum_{i=1}^a \mu_{ij} = a\mu + \underbrace{\sum_i \alpha_i}_0 + a\beta_j + \underbrace{\sum_i (\alpha\beta)_{ij}}_0 \Rightarrow \mu + \beta_j = \frac{1}{a} \sum_i \mu_{ij} = \bar{\mu}_{\bullet j}.$$

Summing over j instead:

$$\mu + \alpha_i = \frac{1}{b} \sum_j \mu_{ij} = \bar{\mu}_{i\bullet}.$$

Summing $\mu + \alpha_i = \bar{\mu}_{i\bullet}$ over i gives $\mu = \bar{\mu}_{\bullet\bullet}$.

Zero-sum parameters are “mean minus mean”

Theorem — parameter meaning under zero-sum constraints

$\mu = \bar{\mu}_{\bullet\bullet}$	overall mean,
$\alpha_i = \bar{\mu}_{i\bullet} - \bar{\mu}_{\bullet\bullet}$	row mean – overall mean,
$\beta_j = \bar{\mu}_{\bullet j} - \bar{\mu}_{\bullet\bullet}$	column mean – overall mean,
$(\alpha\beta)_{ij} = \mu_{ij} - \bar{\mu}_{i\bullet} - \bar{\mu}_{\bullet j} + \bar{\mu}_{\bullet\bullet}$	cell – row – col + overall.

Remark — why we prefer zero-sum for factorial data

These formulas are **symmetric and interpretable** (each effect is a deviation from the grand mean). The models themselves — fitted values, residuals, d.f., SSE — are **identical** whichever constraint is used; only the individual parameter values differ. R reports baseline estimates; `contr.sum` switches to zero-sum.

Parameter Estimation

Least-squares estimates (balanced, zero-sum)

For a balanced design, the estimates under the zero-sum constraints are the sample analogues of the population formulas:

$$\begin{aligned}\hat{\mu}_{ij} &= \bar{y}_{ij\bullet}, & \hat{\mu} &= \bar{y}_{\bullet\bullet\bullet}, \\ \hat{\alpha}_i &= \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}, & \hat{\beta}_j &= \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet}, & \widehat{(\alpha\beta)}_{ij} &= \bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j\bullet} + \bar{y}_{\bullet\bullet\bullet}.\end{aligned}$$

These automatically satisfy the constraints:

$$\sum_i \hat{\alpha}_i = \sum_j \hat{\beta}_j = \sum_i \widehat{(\alpha\beta)}_{ij} = \sum_j \widehat{(\alpha\beta)}_{ij} = 0.$$

- Here $\bar{y}_{i\bullet\bullet}$ = row (A) mean, $\bar{y}_{\bullet j\bullet}$ = column (B) mean, $\bar{y}_{ij\bullet}$ = cell mean, $\bar{y}_{\bullet\bullet\bullet}$ = grand mean.

Balance makes the estimates model-stable

For **any** of the nested reduced models

$$y_{ijk} = \mu + \varepsilon, \quad \mu + \alpha_i + \varepsilon, \quad \mu + \beta_j + \varepsilon, \quad \mu + \alpha_i + \beta_j + \varepsilon,$$

the estimates of μ, α_i, β_j under zero-sum constraints are **identical** to those in the full model:

$$\hat{\mu} = \bar{y}_{\bullet\bullet\bullet}, \quad \hat{\alpha}_i = \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}, \quad \hat{\beta}_j = \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet}.$$

Caution — only under balance

If the design is **unbalanced**, the estimates *change* with the model (just as a regression coefficient changes when other covariates enter). Then the order in which terms are added matters, and Type I sums of squares become ambiguous.

Fitted values under the two models

Full (or means) model: the fitted value is the cell mean,

$$\hat{y}_{ijk} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + \widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij\bullet}$$

Additive model: the fitted value is row + column – overall,

$$\hat{y}_{ijk} = \hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j = \bar{y}_{i\bullet\bullet} + \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet}$$

Remark

Both give the same answer whether one uses baseline or zero-sum coding — the *constraint does not change the fit*. The difference between the two *models* is precisely the interaction array $\widehat{(\alpha\beta)}_{ij}$, which the additive model drops.

Sum-of-Squares Decomposition

From the CRD identity to the factorial split

A balanced $a \times b$ design is also a CRD with ab treatments, so

$$SST = SS_{\text{trt}} + SSE, \quad SS_{\text{trt}} = n \sum_{i,j} (\bar{y}_{ij\bullet} - \bar{y}_{\bullet\bullet\bullet})^2.$$

Because the treatments have factorial structure, SS_{trt} splits further:

$$SS_{\text{trt}} = SS_A + SS_B + SS_{AB}$$

so overall

$$SST = SS_A + SS_B + SS_{AB} + SSE.$$

- d.f. for SS_{trt} is $ab - 1 = (a - 1) + (b - 1) + (a - 1)(b - 1)$ — matching the split.

The sum-of-squares formulas

Source	Sum of squares	d.f.
A	$SS_A = bn \sum_i (\bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = bn \sum_i \hat{\alpha}_i^2$	$a - 1$
B	$SS_B = an \sum_j (\bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet})^2 = an \sum_j \hat{\beta}_j^2$	$b - 1$
AB	$SS_{AB} = n \sum_{i,j} (\bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j\bullet} + \bar{y}_{\bullet\bullet\bullet})^2 = n \sum_{i,j} \widehat{(\alpha\beta)}_{ij}^2$	$(a - 1)(b - 1)$
Error	$SSE = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2$	$ab(n - 1)$

Remark

Each d.f. equals (number of parameters) – (number of constraints): e.g. the interaction has ab parameters with $a + b - 1$ independent constraints, leaving $(a - 1)(b - 1)$.

The error sum of squares

Since each fitted value is the cell mean $\bar{y}_{ij\bullet}$, the residual is

$$e_{ijk} = y_{ijk} - \bar{y}_{ij\bullet},$$

so $\text{SSE} = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij\bullet})^2$ is the **within-cell** variation, pooled over all ab cells.

- ▶ SSE has $ab(n - 1)$ degrees of freedom — $n - 1$ from each of the ab cells.
- ▶ $\text{MSE} = \text{SSE}/[ab(n - 1)]$ is the unbiased estimate of σ^2 .
- ▶ Estimating the error **requires replication** ($n \geq 2$): with $n = 1$ there is no within-cell variation and the interaction cannot be separated from error (recall the RCBD, where additivity was forced for exactly this reason).

ANOVA Table & F-tests

The two-way ANOVA table

Source	d.f.	SS	MS	F
A	$a - 1$	SS_A	$MS_A = \frac{SS_A}{a-1}$	$F_A = \frac{MS_A}{MSE}$
B	$b - 1$	SS_B	$MS_B = \frac{SS_B}{b-1}$	$F_B = \frac{MS_B}{MSE}$
AB	$(a - 1)(b - 1)$	SS_{AB}	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F_{AB} = \frac{MS_{AB}}{MSE}$
Error	$ab(n - 1)$	SSE	$MSE = \frac{SSE}{ab(n-1)}$	
Total	$abn - 1$	SST		

All three F -ratios share the **same denominator** MSE (the within-cell error).

Three questions, three F-tests

1. Does A affect the response?

$$H_0 : \alpha_1 = \cdots = \alpha_a = 0 \Rightarrow F_A = \frac{MS_A}{MSE} \sim F_{a-1, ab(n-1)}.$$

2. Does B affect the response?

$$H_0 : \beta_1 = \cdots = \beta_b = 0 \Rightarrow F_B = \frac{MS_B}{MSE} \sim F_{b-1, ab(n-1)}.$$

3. Do A and B interact?

$$H_0 : (\alpha\beta)_{ij} = 0 \forall i, j \Rightarrow F_{AB} = \frac{MS_{AB}}{MSE} \sim F_{(a-1)(b-1), ab(n-1)}.$$

Remark — test interaction first

Logically the interaction test comes first: if AB is significant, the main-effect tests must be interpreted with care (next section).

Interpreting Main Effects

What does a main-effect null hypothesis really say?

Under zero-sum coding, $\alpha_i = \bar{\mu}_{i\bullet} - \bar{\mu}_{\bullet\bullet}$, so $H_0 : \alpha_1 = \cdots = \alpha_a = 0$ is equivalent to

$$\bar{\mu}_{1\bullet} = \bar{\mu}_{2\bullet} = \cdots = \bar{\mu}_{a\bullet}.$$

That is: the levels of A have the same effect **averaged over the levels of B** .

Caution

This does **not** say the levels of A have the same effect *at each level of B* , i.e. it does *not* imply $\mu_{1j} = \cdots = \mu_{aj}$ for every j — **unless** there is no interaction. When A and B interact, a main-effect average can hide important level-specific differences (or a real effect can average to zero).

Zero main effects, non-zero interaction — a 2×2 example

	$B = 1$	$B = 2$	row mean
$A = 1$	$\mu_{11} = 3$	$\mu_{12} = 2$	$\bar{\mu}_{1\bullet} = 2.5$
$A = 2$	$\mu_{21} = 5$	$\mu_{22} = 0$	$\bar{\mu}_{2\bullet} = 2.5$
col mean	$\bar{\mu}_{\bullet 1} = 4$	$\bar{\mu}_{\bullet 2} = 1$	$\bar{\mu}_{\bullet\bullet} = 2.5$

- ▶ $\alpha_i = \bar{\mu}_{i\bullet} - \bar{\mu}_{\bullet\bullet} = 2.5 - 2.5 = 0$ for both i : **no A main effect**.
- ▶ Yet A clearly **matters**: at $B = 1$, moving A from 1 to 2 raises Y by 2; at $B = 2$, it *lowers* Y by 2.
- ▶ Interaction contrast $= \mu_{11} - \mu_{12} - \mu_{21} + \mu_{22} = 3 - 2 - 5 + 0 = -4 \neq 0$: the lines are not parallel.

Lesson

A non-significant main effect never means “this factor does nothing” when interaction is present. Always read the interaction plot alongside the table.

Worked Example: Sprouting Barley

Marginal and cell means

Cell means $\bar{y}_{ij\bullet}$	1	3	6	9	12	row $\bar{y}_{i\bullet\bullet}$
4 ml	8.67	13.33	21.00	25.33	34.00	20.47
8 ml	4.67	3.67	7.67	6.67	17.00	7.93
col $\bar{y}_{\bullet j\bullet}$	6.67	8.50	14.33	16.00	25.50	14.2

- ▶ Grand mean $\bar{y}_{\bullet\bullet\bullet} = 14.2$.
- ▶ Does water affect germination? The two row means (20.47 vs. 7.93) differ a lot.
- ▶ Does age affect germination? The column means rise steadily $6.67 \rightarrow 25.50$.
- ▶ Do they interact? The gap between the two water rows is *not* constant across ages — something to test.

Estimated effects (zero-sum)

Water main effects $\hat{\alpha}_i = \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}$:

$$\hat{\alpha}_4 = 20.47 - 14.2 = 6.27, \quad \hat{\alpha}_8 = 7.93 - 14.2 = -6.27.$$

Age main effects $\hat{\beta}_j = \bar{y}_{\bullet j \bullet} - \bar{y}_{\bullet\bullet\bullet}$:

$$(-7.53, -5.70, 0.13, 1.80, 11.30) \quad \text{for ages } (1, 3, 6, 9, 12).$$

Interaction array $\widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j \bullet} + \bar{y}_{\bullet\bullet\bullet}$ (row 4 ml):

$$(-4.27, -1.43, 0.40, 3.07, 2.23),$$

and row 8 ml is its negative (zero column sums).

Fitting the model in R

```
barley <- read.table("barley.txt", header = TRUE)
barley$waterfac <- factor(barley$water)
barley$weekfac <- factor(barley$week)

# zero-sum coding, so estimates match the hand calculations
contrasts(barley$waterfac) <- contr.sum(2)
contrasts(barley$weekfac) <- contr.sum(5)

fit <- lm(y ~ waterfac * weekfac, data = barley)
coef(fit)      # (Intercept)=14.2, waterfac1=6.267, weekfac1=-7.533, ...
```

- ▶ `waterfac * weekfac` expands to `waterfac + weekfac + waterfac:weekfac`.
- ▶ The intercept is $\hat{\mu} = 14.2$; `waterfac1` = $\hat{\alpha}_4 = 6.267$; `weekfac1..4` are $\hat{\beta}_1.. \hat{\beta}_4$; the rest follow from the zero-sum constraints.

The ANOVA table

```
anova(fit)
```

```
Response: y
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
waterfac	1	1178.1	1178.1	19.723	0.00025
weekfac	4	1321.1	330.3	5.529	0.00364
waterfac:weekfac	4	208.9	52.2	0.874	0.49673
Residuals	20	1194.7	59.7		

- ▶ Water: $F_A = 19.7$, $p = 0.00025$ — **strong** main effect.
- ▶ Age: $F_B = 5.53$, $p = 0.0036$ — **significant** main effect.
- ▶ Interaction: $F_{AB} = 0.87$, $p = 0.50$ — **no** evidence of interaction.

Caution

Do **not** interpret yet — we must check the model assumptions first.

Diagnostics & Transformation

Checking assumptions (1) — residuals

```
plot(fitted(fit), resid(fit), xlab = "Fitted Values", ylab = "Residuals")  
abline(h = 0, lty = 2)  
qqnorm(resid(fit)); qqline(resid(fit))
```

- ▶ Residual-vs-fitted plot: the spread of the noise grows with the fitted values (a funnel shape) — a sign of non-constant variance.
- ▶ Normal Q–Q plot: the tails are heavy (points bend away from the line).

Remark — why counts misbehave

The response is a *count* of germinated seeds. Counts typically have variance that grows with the mean, so a variance-stabilizing transformation is natural.

Checking assumptions (2) — Box–Cox

```
library(MASS)
boxcox(fit)           # profile log-likelihood for the power lambda
```

- ▶ The 95% interval for λ includes $1/2 \Rightarrow$ take the square root of the response.
- ▶ Refit on the new scale:

```
fit2 ← lm(sqrt(y) ~ waterfac * weekfac, data = barley)
```

Caution

A transformation changes the model — so we must re-check the assumptions of `fit2` before trusting its ANOVA table.

Re-check after the square-root transform

```
plot(fitted(fit2), resid(fit2)); abline(h = 0, lty = 2)
qqnorm(resid(fit2)); qqline(resid(fit2))
boxcox(fit2)
```

- ▶ Residual spread now looks **constant** across fitted values.
- ▶ The Q–Q plot shows **no** obvious non-normality.
- ▶ The Box–Cox 95% interval for λ now **includes 1**: the current (square-root) scale is fine — no further transformation needed.

Remark

Diagnostics drove the analysis: we transformed *because* the plots demanded it, not by habit. Only now are the p -values trustworthy.

ANOVA after transformation

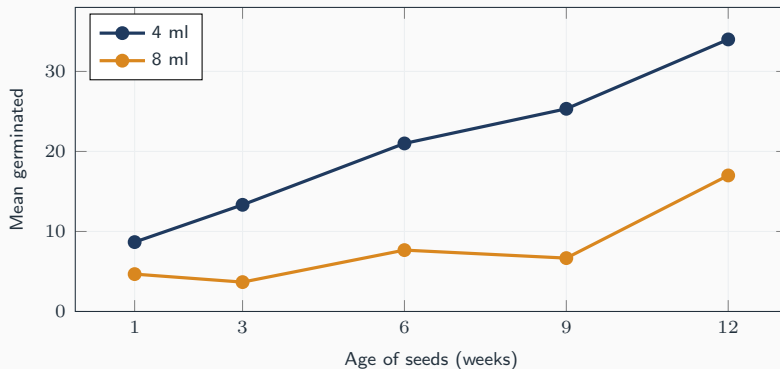
```
anova(fit2)
```

```
Response: sqrt(y)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
waterfac	1	21.89	21.893	23.761	0.000092
weekfac	4	21.89	5.474	5.941	0.002555
waterfac:weekfac	4	2.25	0.562	0.610	0.660139
Residuals	20	18.43	0.921		

- ▶ The conclusions are **qualitatively unchanged and now reliable**: both main effects are significant (water $p \approx 0.0001$, age $p \approx 0.0026$), interaction is not ($p \approx 0.66$).
- ▶ Water and age each affect germination; their effects are **additive** on the square-root scale.

Interaction plot for the barley data



- ▶ The 4 ml line sits **above** the 8 ml line (water main effect); both **rise** with age (age main effect).
- ▶ The lines are **roughly parallel** — consistent with the non-significant interaction test.

Factorial vs. One-at-a-Time

Factorial vs. one-at-a-time experiments

To study two factors, one *could* run two separate experiments, varying one factor at a time (holding the other fixed). Compare that with a single factorial design that crosses every level of A with every level of B .

Remark — the factorial wins on three counts

A factorial design is superior to one-at-a-time because it can:

- ▶ test **both** factors at once — more *efficient*, fewer units for the same precision (each observation helps estimate *both* main effects);
- ▶ estimate the **interaction** — one-at-a-time designs are structurally blind to it;
- ▶ **broaden the inductive base**: conclusions hold across a range of levels of the other factor, not just one fixed setting.

Summary

1. A two-way factorial design crosses a levels of A with b levels of B ; it is a CRD with ab treatments plus **factorial structure**.
2. **Interaction** = the effect of one factor depends on the level of the other; parallel lines in an interaction plot mean *no* interaction.
3. Full model $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon$ is over-parameterized $\Rightarrow$ needs $1 + a + b$ constraints (zero-sum or baseline).
4. Balanced estimates: $\hat{\alpha}_i = \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet\bullet\bullet}$, $\hat{\beta}_j = \bar{y}_{\bullet j\bullet} - \bar{y}_{\bullet\bullet\bullet}$, $\widehat{(\alpha\beta)}_{ij} = \bar{y}_{ij\bullet} - \bar{y}_{i\bullet\bullet} - \bar{y}_{\bullet j\bullet} + \bar{y}_{\bullet\bullet\bullet}$.
5. $SST = SS_A + SS_B + SS_{AB} + SSE$; all three F -tests use MSE; error needs replication ($n \geq 2$).
6. **Read interaction first**; a main effect is an average over the other factor and can mislead when interaction is present.
7. **Check assumptions** (residuals, Q-Q, Box-Cox) before interpreting; counts often need a square-root scale.

Remark

- ▶ With $n = 1$ (no replication) the interaction cannot be separated from error — which is exactly the **RCBD** situation, where additivity was assumed by necessity.
- ▶ When one factor is a nuisance grouping rather than of interest, the same machinery becomes **blocking**.
- ▶ If the normality assumption is doubtful, permute observations across cells to test main effects and interaction non-parametrically.
- ▶ Natural extensions: three-way and higher factorials, fractional factorials, and random-effects / mixed models.